

United International University
School of Business and Economics
Real-Life Project: NLP & Sentiment Analysis

Online Review Sentiment Analysis — Flipkart / Twitter

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Report: Project Summary & Result Interpretation

Project brief

Building a complete sentiment analysis pipeline on a dataset of online product reviews and social media posts (a synthetic Flipkart / Twitter-style dataset). The pipeline consisting of raw text, clean; also vectorized. It's been classified into positive, negative, or neutral sentiment. There have been described the findings of the project as well.

LOAD of Dataset

A given reference file of CSV format on "online_reviews_sentiment" from source (Flipkart / Twitter) imported/loaded & basic EDA & data preprocessing, TF-IDF, model training, evaluation, and trend visualization done.

Model Selection, Implementation, Evaluation and Model Comparison

Building the end-to-end pipeline turns the raw review text into a trained sentiment classifier; evaluated each trained model fairly on the same held-out test set, and compared them.

The worked out IPYNB file is given in attachment.

Result Interpretation and Report

(1) Pipeline Summary and Model Comparison

The sentiment analysis pipeline was constructed to ingest, clean, and classify raw online review text. The preprocessing phase standardized the text by converting it to lowercase, stripping out URLs, mentions, punctuation, and standalone numbers, and removing common English stopwords. To improve upon standard tokenization, a lemmatizer was applied to reduce words, ensuring cleaner feature extraction. The text was then vectorized using TF-IDF, specifically tuned to ignore highly rare words and overly common terms to reduce noise.

Three classification models were trained on a stratified 80/20 split: a class-weighted Logistic Regression, a Random Forest, and a Gradient Boosting classifier. The evaluation revealed that all three models achieved a perfect 1.0 F1-score. Because all models performed flawlessly, Logistic Regression was selected as the optimal model due to its high computational efficiency, straightforward interpretability, and ability to output calibrated probabilities.

(2) Sentiment Trends Analysis

- **Overall Distribution:** The dataset exhibits a relatively balanced distribution, comprising 400 positive reviews, 340 negative reviews, and 260 neutral reviews. This ensures the model receives sufficient examples of each class without severe imbalance.
- **Sentiment Over Time:** Monthly trend analysis shows a stable, parallel trajectory across all three sentiment classes throughout the year. There are no extreme seasonal spikes or drops, which suggests a consistent sampling of opinions rather than a dataset skewed by a specific viral event or holiday season.
- **Sentiment by Source:** There is a distinct behavioral split between the two platforms. Flipkart reviews lean predominantly positive and neutral, reflecting standard post-purchase product evaluations. Conversely, Twitter data contains a noticeably higher concentration of negative sentiment, highlighting its common real-world use as an escalation channel for frustrated customers seeking immediate support.

(3) Confusion Matrix Interpretation

The confusion matrix for the test set (comprising 68 negative, 52 neutral, and 80 positive instances) shows zero misclassifications. Because this is a synthetic dataset generated for teaching purposes, the text relies on distinct, non-overlapping templates that a TF-IDF vectorizer can easily separate.

However, in a real-world scenario, **neutral sentiment** is historically the hardest to classify correctly. Neutral reviews often contain a mix of both positive and negative

vocabulary (e.g., "The battery life is great, but the screen is too dim") or lack strong polarizing adjectives altogether. Models frequently confuse these mixed reviews with either slight positive or slight negative classes, depending on which specific keywords carry the highest TF-IDF weights.

(4) Limitations and Future Improvements

- **Limitation 1: Synthetic Data and Lack of Nuance:** Because the dataset is synthetic, it lacks the chaotic nature of authentic human text. Real reviews contain sarcasm, slang, misspellings, and colloquialisms that easily break simple bag-of-words models. A perfect 1.0 F1-score strongly indicates the model has memorized structural templates rather than learned true linguistic sentiment.
- **Limitation 2: TF-IDF Ignores Word Order:** The TF-IDF approach treats text as an unordered collection of words and bigrams. It cannot understand deep semantic context, meaning phrases like "not good" and "good, not bad" can easily confuse the model if the n-gram range does not perfectly capture the negation.
- **Concrete Improvement:** To improve the model for real-world application, the TF-IDF vectorizer should be replaced with pre-trained contextual embeddings, such as a transformer model (e.g., BERT or RoBERTa). Transformers process words in relation to all other words in a sentence, allowing the algorithm to understand complex context, sarcasm, and negation accurately.

(5) Business Implications

Deploying this sentiment analysis pipeline allows a business to automatically triage incoming customer feedback at scale. Because the data shows Twitter is primarily used for negative complaints while Flipkart captures positive product experiences, the company should optimize its workflow by platform. Twitter integrations should immediately route negative sentiments to an urgent customer retention and crisis-management team. Meanwhile, the predominantly positive Flipkart reviews can be automatically aggregated by the marketing department to identify successful product features and generate promotional testimonials.